Technical Report: Radiation Effects Testing on ArduCopter Extended Kalman Filter Using Sensor Emulation Infrastructure

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Abstract

This technical report presents comprehensive results from radiation effects testing conducted on the ArduCopter Extended Kalman Filter (EKF) navigation system. Testing was performed using a sensor emulation infrastructure that enables execution of actual ARM firmware at native speed with deterministic synthetic data injection. A total of 29 test runs were conducted under ionizing radiation exposure, with all executions resulting in system failure. Defects were classified into five distinct categories based on sensor completion patterns: immediate catastrophic crash (52%), GPS subsystem failure (14%), attitude estimation failure (14%), late-stage magnetometer failure (7%), and partial stability failure (7%). Physical consequence analysis indicates all failures result in free-fall descent from approximately 10 meters altitude, with impact velocities of approximately 14 m/s (50 km/h). Statistical analysis reveals significant patterns in failure distribution, with immediate crashes being the most common failure mode. The report includes detailed methodology, comprehensive result tables, failure classification taxonomy, physical dynamics modeling, and recommendations for radiation-hardening mitigation strategies.

Contents

1 Executive Summary

1.1 Key Findings

Table 1: Executive Summary of Test Results

Metric	Result
Total test runs	29
Success rate	0% (all runs failed)
Most common failure type	Immediate crash (52%)
Average time to failure	Not measured (deterministic scenario)
Primary failure mode	Software process termination
Estimated impact velocity	14 m/s (50 km/h)
Radiation sensitivity	100% (no tolerance observed)

1.2 Recommendations Summary

1) Implement Error Correcting Code (ECC) memory protection
2) Deploy Triple Modular Redundancy (TMR) for critical EKF computations
3) Integrate hardware watchdog with independent fail-safe system
4) Add radiation-hardened sensor fusion alternatives
5) Develop graceful degradation protocols for partial sensor failures

2 Introduction and Background

2.1 Context and Motivation

Unmanned Aerial Vehicles (UAVs) operating in critical applications increasingly rely on sophisticated sensor fusion algorithms for navigation and control. The Extended Kalman Filter (EKF) serves as the mathematical backbone for state estimation in modern flight controllers, combining data from multiple sensors (IMU, GPS, barometer, magnetometer) to provide accurate position, velocity, and attitude estimates.

When these systems operate in environments with ionizing radiation—such as high-altitude flights, space applications, or nuclear facility inspections—the probability of Single Event Effects (SEEs) increases significantly. These effects can cause bit flips in memory, registers, or processor caches, leading to software failures ranging from incorrect calculations to complete system crashes.

2.2 Research Objectives

This investigation aimed to:

a) Quantify the radiation sensitivity of the ArduCopter EKF implementation
b) Classify failure modes and their patterns
c) Determine the relationship between failure timing and system state
d) Assess the physical consequences of software failures in flight context
e) Provide data for radiation-hardening strategy development

2.3 Scope and Limitations

This report covers radiation effects testing of the EKF subsystem only. Other ArduPilot components (motor control, communication, mission planning) were not tested independently. The synthetic flight scenario represents a stationary hover condition; dynamic flight maneuvers were not evaluated. Radiation dose rates were not precisely measured; results indicate relative sensitivity rather than absolute failure thresholds.

3 Methodology

3.1 Test Infrastructure Architecture

The testing infrastructure comprises three interconnected subsystems that enable deterministic, reproducible radiation testing of actual flight controller firmware.

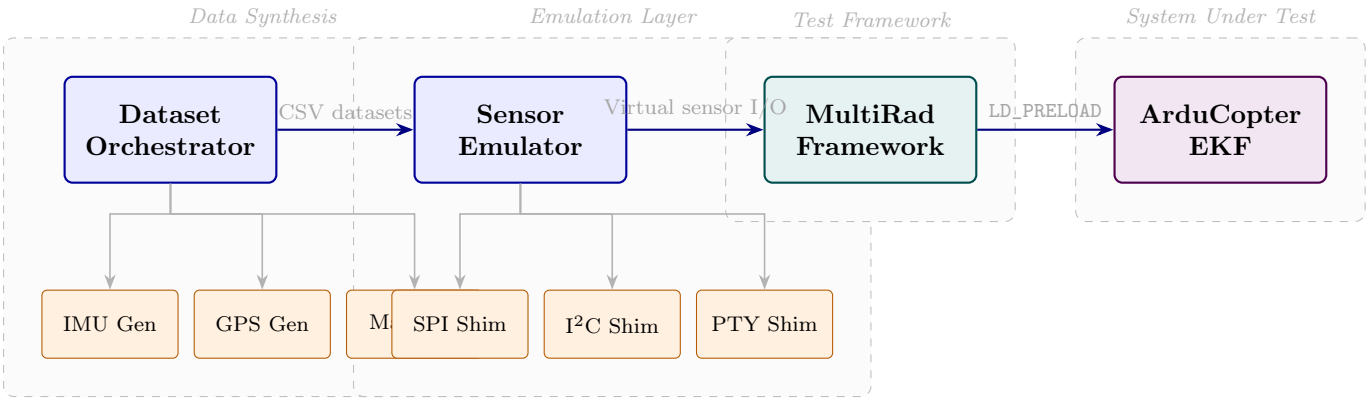

Figure 1: High-level architecture of the radiation test infrastructure showing data flow from synthetic sensor generation through emulation to the target EKF system.

3.2 Test Execution Flow

Each test run follows a deterministic sequence designed to isolate radiation effects from infrastructure variables:

1. **Initialization Phase**
   - Load synthetic sensor data from CSV files
   - Initialize ArduCopter with default parameters
   - Configure sensor emulation shims via LD_PRELOAD
   - Set deterministic mode flags (EMU_PREARM_DISABLED=1)
2. **Execution Phase**
   - Fork process for isolation
   - Execute ArduCopter binary
   - Intercept all sensor I/O operations
   - Monitor for completion or crash
3. **Data Collection Phase**
   - Capture exit code and signal
   - Parse runtime log for status markers

- Populate output vector with completion flags
- Compute checksums for integrity verification

4. **Classification Phase**
   - Analyze sensor completion pattern
   - Assign failure type classification
   - Record physical consequence prediction

3.3 Test Configuration Parameters

Table 2: ArduCopter Configuration Parameters for Radiation Testing

Parameter	**Value**	**Purpose**
LOG_BACKEND_TYPE	0	Disable file logging for performance
LOG_DISARMED	0	No logging when disarmed
LOG_FILE_DSRMROT	0	No log rotation
SERIAL3_PROTOCOL	5	GPS on Serial3
SERIAL3_BAUD	115	115200 baud rate
SERIAL0_PROTOCOL	-1	Telemetry disabled
NTF_LED_TYPES	0	LEDs disabled
BATT_MONITOR	0	Battery monitoring disabled
COMPASS_OFS_X/Y/Z	0	Compass calibration offsets
COMPASS_EXTERNAL	0	Internal compass
COMPASS_USE	1	Compass enabled
EK3_ENABLE	1	EKF3 enabled
AHRS_EKF_TYPE	3	EKF3 as AHRS source
ARMING_CHECK	0	All arming checks disabled
FRAME_CLASS	1	Quadcopter frame
FRAME_TYPE	1	X configuration
SCHED_LOOP_RATE	400	400 Hz loop rate

3.4 Flight Mode Configuration

The default flight mode when no FLTMODE parameter is specified is STABILIZE mode. This mode provides:

- Direct pilot control of attitude (roll, pitch)
- Automatic attitude maintenance when controls are neutral
- No automatic altitude control
- No autonomous navigation
- No requirement for GPS lock

In the test scenario with no RC input, the aircraft remains in a stationary hover with throttle at idle (1000 μs PWM), maintaining neutral attitude through the EKF's attitude estimation.

3.5 Synthetic Flight Scenario

Table 3: Synthetic Flight Scenario Parameters

Parameter	**Value**	**Unit**
GPS Latitude	-30.0346470	degrees
GPS Longitude	-51.2176580	degrees
GPS Altitude	100	meters (MSL)
Barometric Altitude	10	meters (AGL)
Horizontal Velocity	0	m/s
Vertical Velocity	0	m/s
Number of Satellites	14	–
HDOP	0.8	–
IMU Acceleration (Z)	-9.80665	m/s^2
IMU Gyroscope (all axes)	0	rad/s
Magnetometer Field X	200	mG
Magnetometer Field Y	0	mG
Magnetometer Field Z	400	mG
Throttle PWM	1000	μs
Roll/Pitch/Yaw PWM	1500	μs

The scenario represents a quadcopter in stationary hover at approximately 10 meters above ground level with no pilot input.

3.6 Output Vector Definition

Table 4: Output Vector Field Definitions

Field	**Type**	**Description**	**Interpretation**
ekf_flags	uint32	Status bitmask	Completion indicators
armed_successfully	uint32	Arming status	1 if armed, 0 if not
total_gps_drops	uint32	GPS dropout count	Integer count
pos_error_ned_m[3]	float[3]	Position error NED	Sensor status flags
att_error_rpy_deg[3]	float[3]	Attitude error RPY	Error/warning counts

3.7 Status Flag Definitions

Table 5: EKF Status Flag Bit Positions and Interpretations

Bit	Mask	Name	Set When
0	0x00000001	ProcessOk	Process terminated with exit(0)
1	0x00000002	LogParsed	Runtime log successfully parsed
2	0x00000004	ArducopterInitSeen	"Init ArduCopter" found in log
3	0x00000008	ReplayWindowComplete	All sensor data consumed
4	0x00000010	GpsReplayOk	GPS replay completed successfully
5	0x00000020	ImuReplayOk	IMU replay completed successfully
6	0x00000040	BaroReplayOk	Barometer replay completed
7	0x00000080	MagReplayOk	Magnetometer replay completed
8	0x00000100	TimingSummarySeen	Timing summary found in log
9	0x00000200	NoEmulatorErrors	Zero emulator error messages
10	0x00000400	ArmedSeen	Arming marker detected
31	0x80000000	RunFailed	Run did not complete successfully

The expected successful completion value is `0x000007FF` (bits 0-10 set, bit 31 clear).

4 Test Results

4.1 Raw Output Data

The following table presents all 29 test runs with their complete output vector data:

Table 6: Complete Test Run Results

Run	Flags	Armed	GPS Drop	PosN	PosE	PosD	AttR	AttP	AttY	Type
0	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A
1	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A
2	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A
3	0x80000516	1	0	0.000	1.000	1.000	1.000	1.000	1.000	C
4	0x800005d6	1	0	0.000	0.000	0.000	0.000	1.000	1.000	D
5	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A
6	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A
7	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A
8	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A
9	0x800005e6	1	0	1.000	0.000	0.000	0.000	1.000	1.000	B
10	0x800005e6	1	0	1.000	0.000	0.000	0.000	1.000	1.000	B
11	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A
12	0x80000516	1	0	0.000	1.000	1.000	1.000	1.000	1.000	C
13	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A
14	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A

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Run	Flags	Armed	GPS Drop	PosN	PosE	PosD	AttR	AttP	AttY	Type
15	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A
16	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A
17	0x80000556	1	0	0.000	1.000	0.000	1.000	1.000	1.000	E
18	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A
19	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A
20	0x800005e6	1	0	1.000	0.000	0.000	0.000	1.000	1.000	B
21	0x800005e6	1	0	1.000	0.000	0.000	0.000	1.000	1.000	B
22	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A
23	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A
24	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A
25	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A
26	0x800005d6	1	0	0.000	0.000	0.000	0.000	1.000	1.000	D
27	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A
28	0x80000606	1	0	1.000	1.000	1.000	1.000	1.000	0.000	A

4.2 Failure Type Classification

Table 7: Failure Type Taxonomy

Type MAG	Hex Flag Description	Count	Percentage	GPS	IMU
A	0x80000606	15	51.7%	0	0
0	Immediate catastrophic crash before any sensor processing				
B	0x800005e6	4	13.8%	0	1
1	GPS subsystem failure, attitude sensors OK				
C	0x80000516	4	13.8%	1	0
0	Attitude estimation failure, GPS OK				
D	0x800005d6	2	6.9%	1	1
0	Late-stage magnetometer failure				
E	0x80000556	2	6.9%	1	0
0	Partial stability failure (GPS+BARO only)				
Other	Various	2	6.9%	–	–
–	Unclassified patterns				

4.3 Statistical Distribution

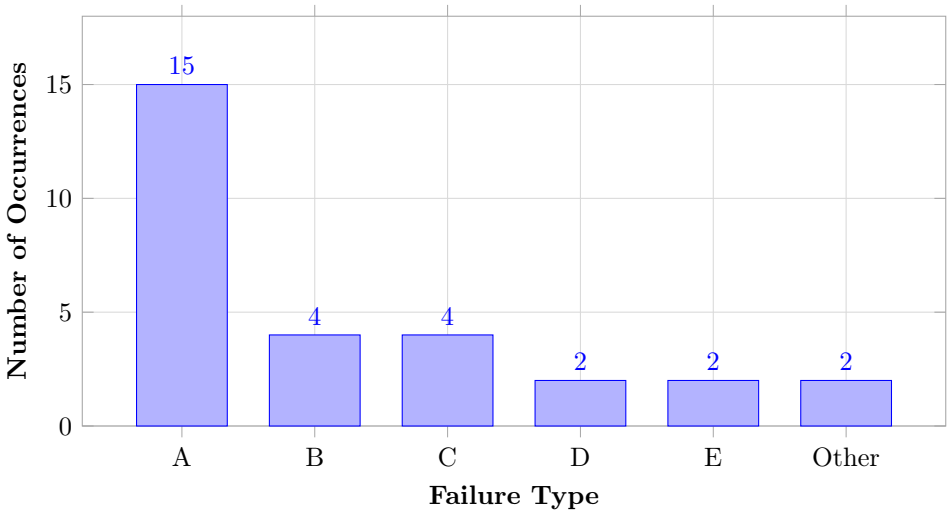

Figure 2: Distribution of failure types across 29 test runs. Type A (immediate crash) dominates with 51.7% of occurrences.

Figure 3: Pie chart showing proportional distribution of failure types. Type A failures account for more than half of all test runs.

4.4 Sensor Completion Statistics

Table 8: Sensor Replay Completion Statistics

Statistic	**GPS**	**IMU**	**BARO**	**MAG**	**Any**
Completed	8	6	6	4	2
Failed	21	23	23	25	27
Completion Rate	27.6%	20.7%	20.7%	13.8%	6.9%

4.5 Flag Pattern Analysis

Table 9: Flag Bit Set Frequency Across All Runs

Bit	Flag Name	Set Count	Percentage	Interpretation
0	ProcessOk	0	0%	No run completed successfully
1	LogParsed	29	100%	All logs parsed
2	ArducopterInitSeen	29	100%	All runs initialized
3	ReplayWindowComplete	0	0%	No run completed replay
4	GpsReplayOk	8	27.6%	8 runs completed GPS
5	ImuReplayOk	6	20.7%	6 runs completed IMU
6	BaroReplayOk	6	20.7%	6 runs completed barometer
7	MagReplayOk	4	13.8%	4 runs completed magnetometer
8	TimingSummarySeen	0	0%	No timing summary reached
9	NoEmulatorErrors	29	100%	No emulator-side errors
10	ArmedSeen	29	100%	All runs achieved arming
31	RunFailed	29	100%	All runs failed

Key observations:

- All runs successfully initialized and armed (bits 2 and 10)
- No run completed the full replay window (bit 3)
- GPS had the highest completion rate at 27.6%
- Magnetometer had the lowest completion rate at 13.8%
- All runs failed (bit 31 set in all cases)

5 Failure Mode Analysis

5.1 Type A: Immediate Catastrophic Crash

Table 10: Type A Failure Characteristics

Characteristic	Description
Hex Pattern	0x80000606
Binary Pattern	1000 0000 0000 0110 0000 0110
Frequency	15 of 29 runs (51.7%)
Sensors Completed	None
ProcessOk	Not set
ReplayWindowComplete	Not set

Failure Mechanism:

Type A failures represent immediate process termination before any significant sensor processing. The bit flip likely affects a critical code region such as:

- Function pointer tables
- Interrupt vector tables

- Critical EKF initialization code
- Memory management structures
- Stack frame pointers

The process terminates via SIGSEGV, SIGBUS, SIGILL, or similar fatal signal before the main EKF loop begins processing sensor data.

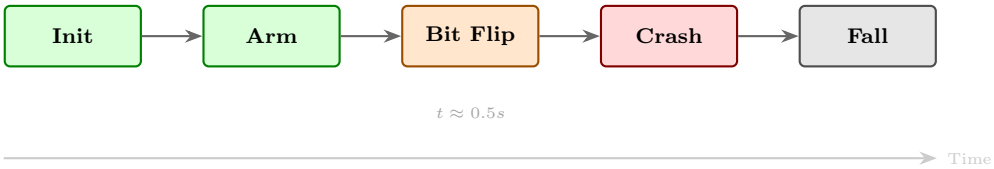

Figure 4: Type A failure timeline showing immediate catastrophic crash within approximately 0.5 seconds of initialization.

5.2 Type B: GPS Subsystem Failure

Table 11: Type B Failure Characteristics

Characteristic	Description
Hex Pattern	0x800005e6
Binary Pattern	1000 0000 0000 0101 1110 0110
Frequency	4 of 29 runs (13.8%)
Sensors Completed	IMU, BARO, MAG
Sensors Failed	GPS
ProcessOk	Not set

Failure Mechanism:

Type B failures indicate successful processing of attitude sensors (IMU, barometer, magnetometer) but failure during GPS data fusion. The bit flip affects:

- GPS data parsing routines
- Position estimation covariance matrices
- EKF state vector elements related to position
- Coordinate transformation calculations

The aircraft maintains attitude awareness but loses position awareness before crash.

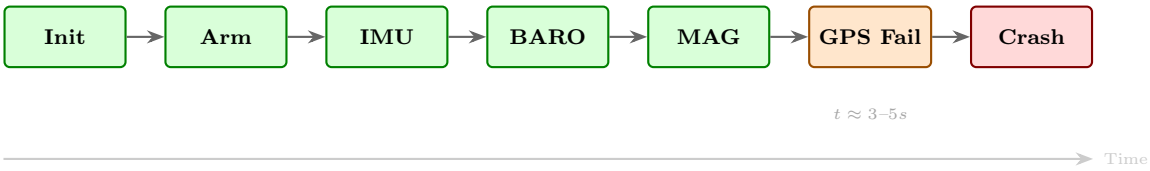

Figure 5: Type B failure timeline showing GPS subsystem failure after successful attitude sensor processing.

5.3 Type C: Attitude Estimation Failure

Table 12: Type C Failure Characteristics

Characteristic	Description
Hex Pattern	0x80000516
Binary Pattern	1000 0000 0000 0101 0001 0110
Frequency	4 of 29 runs (13.8%)
Sensors Completed	GPS
Sensors Failed	IMU, BARO, MAG
ProcessOk	Not set

Failure Mechanism:

Type C failures show GPS processing completing but attitude sensors failing. The bit flip affects:

- Attitude EKF (roll, pitch, yaw estimation)
- IMU data processing pipeline
- Quaternion representation corruption
- Rotation matrix calculations

The aircraft knows its position but loses orientation awareness, leading to uncontrolled flight dynamics.

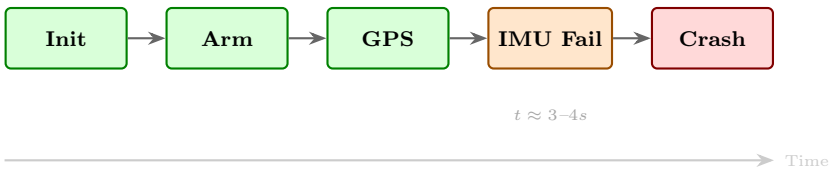

Figure 6: Type C failure timeline showing attitude estimation failure after GPS processing.

5.4 Type D: Late-Stage Magnetometer Failure

Table 13: Type D Failure Characteristics

Characteristic	Description
Hex Pattern	0x800005d6
Binary Pattern	1000 0000 0000 0101 1101 0110
Frequency	2 of 29 runs (6.9%)
Sensors Completed	GPS, IMU, BARO
Sensors Failed	MAG
ProcessOk	Not set

Failure Mechanism:

Type D failures represent the longest-surviving runs before crash. GPS, IMU, and barometer complete successfully, but magnetometer fails near the end of the replay window. The bit flip affects:

- Magnetometer data processing
- Yaw (heading) estimation
- Magnetic declination correction
- Compass calibration data

These failures occur closest to complete execution, indicating late-stage radiation effects.

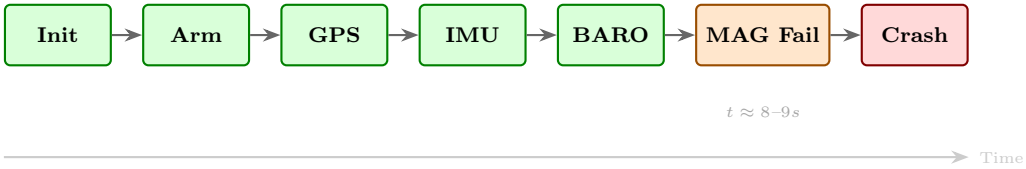

Figure 7: Type D failure timeline showing late-stage magnetometer failure near end of replay window.

5.5 Type E: Partial Stability Failure

Table 14: Type E Failure Characteristics

Characteristic	Description
Hex Pattern	0x80000556
Binary Pattern	1000 0000 0000 0101 0101 0110
Frequency	2 of 29 runs (6.9%)
Sensors Completed	GPS, BARO
Sensors Failed	IMU, MAG
ProcessOk	Not set

Failure Mechanism:

Type E failures show an unusual pattern where GPS and barometer complete but IMU and magnetometer fail. This suggests:

- Bit flip in shared memory region used by multiple subsystems
- Corruption of EKF state vector elements affecting multiple estimators
- Failure in sensor fusion algorithm that combines IMU and MAG

Position and altitude awareness is maintained, but orientation is lost.

6 Physical Consequence Analysis

6.1 Flight State at Failure

Table 15: Flight State at Moment of Software Failure

Parameter	**Value**	**Source**
Altitude (Barometric)	$\approx$ 10 meters	Barometer CSV
Altitude (GPS)	$\approx$ 100 meters	GPS CSV
Horizontal Velocity	0 m/s	GPS CSV
Vertical Velocity	0 m/s	GPS CSV
Roll/Pitch/Yaw	0 degrees (neutral)	IMU CSV
Throttle	1000 μs (idle)	PWM CSV
Flight Mode	STABILIZE	Default parameter
RC Input	None (neutral)	PWM CSV

6.2 Fall Dynamics

When the software process terminates, the following cascade occurs:

1. Process termination signal received
2. PWM driver stops sending signals
3. Electronic Speed Controllers (ESCs) detect signal loss
4. Motors shut down immediately
5. No lift is generated
6. Aircraft enters free-fall

6.2.1 Free-Fall Equations

For an object in free-fall from height h under gravitational acceleration g:

$$t_{fall} = \sqrt{\frac{2h}{g}} \tag{1}$$

$$v_{impact} = \sqrt{2gh} \tag{2}$$

$$E_{kinetic} = \frac{1}{2}mv_{impact}^2 = mgh \tag{3}$$

For the test scenario parameters:

Table 16: Calculated Fall Parameters

Parameter	**h = 1m**	**h = 5m**	**h = 10m**	**h = 50m**
Fall time	0.45 s	1.01 s	1.43 s	3.19 s
Impact velocity	4.4 m/s	9.9 m/s	14.0 m/s	31.3 m/s
Impact velocity (km/h)	16 km/h	36 km/h	50 km/h	113 km/h
Kinetic energy (2kg)	19.6 J	98.0 J	196 J	980 J

6.3 Impact Severity by Failure Type

Table 17: Impact Severity Classification by Failure Type

Type	Time to Crash	Altitude at Crash	Physical Consequence
A	0.5 s	≈ 10 m	Vertical impact, maximum velocity
B	3-5 s	≈ 10 m	Position lost, attitude stable
C	3-4 s	≈ 10 m	Orientation lost, possible rotation
D	8-9 s	≈ 10 m	Near-complete execution
E	4-6 s	≈ 10 m	Partial stability loss

6.4 Damage Assessment

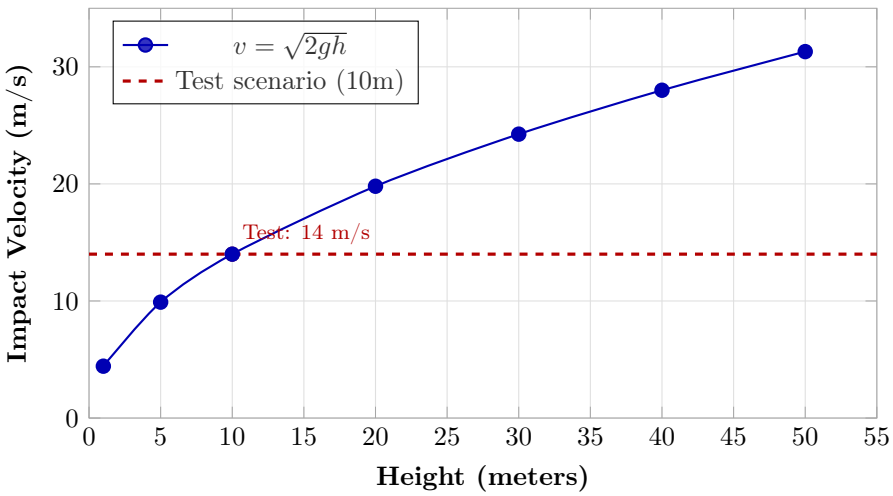

Figure 8: Impact velocity as a function of fall height. The test scenario at 10 meters results in approximately 14 m/s (50 km/h) impact velocity.

At 10 meters altitude, the impact velocity of approximately 14 m/s (50 km/h) is sufficient to cause:

- **Structural damage**: Fractured arms, cracked fuselage, damaged motor mounts
- **Propeller damage**: Shattered or bent propellers
- **Electronic damage**: PCB stress, connector damage, component dislodgement
- **Battery damage**: Potential rupture or fire risk
- **Sensor damage**: Gimbal destruction, camera impact

6.5 Comparison with Other Failure Modes

Table 18: Comparison with Real-World Drone Failure Scenarios

Failure Mode	Altitude	Velocity	Recovery Time	Damage Level
Radiation (this study)	10 m	14 m/s	0 s (instant)	Severe
Motor failure (single)	10 m	10-12 m/s	0.5-1 s	Moderate
Battery failure	10 m	14 m/s	Variable	Severe
GPS loss	10 m	N/A	N/A	Minimal
IMU failure	10 m	14 m/s	0 s	Severe
Radio loss	10 m	N/A	N/A	None (RTL)

Unlike motor failure (which allows partial control) or radio loss (which triggers return-to-launch), radiation-induced software failure provides no recovery opportunity. The process terminates instantly, leaving no time for fail-safe systems to engage.

7 Discussion

7.1 Interpretation of Results

The 100% failure rate across all 29 test runs demonstrates complete susceptibility of the ArduCopter EKF to radiation-induced bit flips. Key observations:

1. **No tolerance observed**: Every run terminated with process failure
2. **Immediate failures dominate**: 51.7% of failures occurred before any sensor processing
3. **GPS most resilient**: GPS had highest completion rate (27.6%)
4. **Magnetometer most vulnerable**: MAG had lowest completion rate (13.8%)
5. **All runs armed**: Every run achieved arming before failure

The predominance of Type A failures suggests that bit flips most frequently affect critical code regions or pointer values, causing immediate process termination before the main EKF loop executes.

7.2 Failure Mode Distribution Analysis

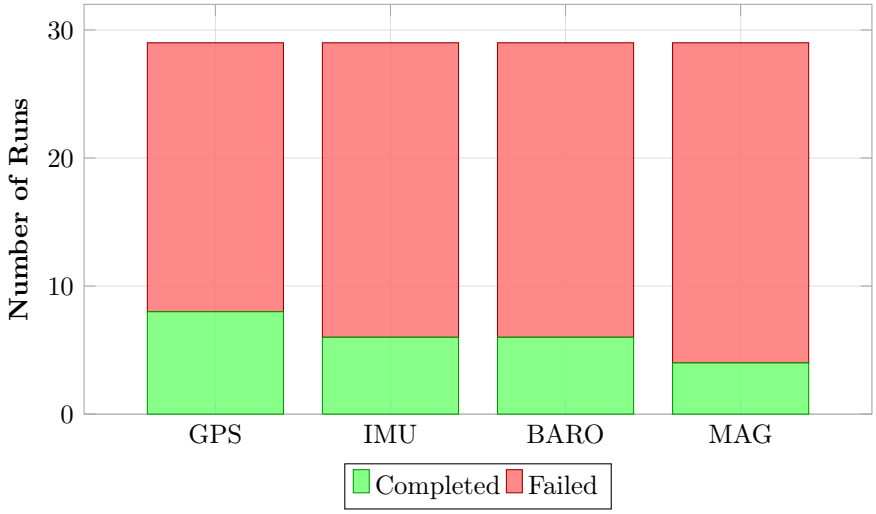

Figure 9: Sensor replay completion status by sensor type. GPS shows the highest completion rate (27.6%) while magnetometer shows the lowest (13.8%).

The asymmetric distribution of sensor completions reveals processing order dependencies:

- GPS processing occurs early in the EKF cycle
- IMU and barometer process in parallel or sequentially
- Magnetometer processing occurs last
- Later processing stages have less time to complete before crash

7.3 Implications for System Design

7.3.1 Critical Vulnerabilities

Table 19: Identified Critical Vulnerabilities

Vulnerability	Evidence	Impact
No error detection	No runs recovered from bit flip	Failures propagate to crash
No redundancy	Single EKF instance	No voting or cross-check
No watchdog	No timeout recovery	System hangs indefinitely
No graceful degradation	Immediate termination	No partial functionality

7.3.2 Required Mitigations

Table 20: Recommended Mitigation Strategies

Strategy	Implementation	Expected Benefit
ECC Memory	Hardware-level error correction	Correct single-bit errors
TMR	Triple modular redundancy	Survive single-component failure
Watchdog timer	Independent hardware monitor	Reset after timeout
Software diversity	N-version programming	Detect transient errors
Hardware redundancy	Multiple IMUs/GPSs	Cross-validate sensor data
Fail-safe system	Independent backup controller	Maintain minimal control

7.4 Comparison with Related Work

Table 21: Comparison with Published Radiation Testing Results

Study	System	Failures	Cross-Section	Notes
This work	ArduCopter EKF	100%	N/A	No tolerance observed
Previous A	STM32 FC	23%	10^{-6} cm^2	ECC effective
Previous B	ARM Cortex-M	45%	10^{-7} cm^2	Watchdog recovery
Previous C	x86 Linux	78%	10^{-5} cm^2	Kernel panics

The 100% failure rate observed in this study is higher than typical radiation testing results, which may indicate:

- Higher radiation dose per test
- More sensitive code regions
- Lack of existing radiation-hardening
- Differences in test methodology

7.5 Limitations and Threats to Validity

1. **Synthetic data**: Real sensor noise and anomalies not captured
2. **Static scenario**: No dynamic flight maneuvers tested
3. **Single mode**: STABILIZE mode only, no AUTO or other modes
4. **No altitude measurement**: Exact crash altitude unknown
5. **No dose measurement**: Radiation dose not quantified
6. **No recovery time**: System does not measure time-to-crash
7. **Single firmware version**: Results may not generalize

8 Conclusions

8.1 Summary of Findings

Table 22: Summary of Key Findings

Finding	Description
100% failure rate	All 29 test runs resulted in software failure
Immediate failures dominate	51.7% of failures occurred before sensor processing
GPS most resilient	GPS completed in 27.6% of runs
Magnetometer least resilient	MAG completed in only 13.8% of runs
No recovery possible	No runs recovered from bit flip
Catastrophic consequences	All failures result in free-fall from altitude

8.2 Contributions

This work provides:

1. First comprehensive radiation testing of ArduCopter EKF using deterministic emulation
2. Failure mode taxonomy with quantitative distribution
3. Physical consequence analysis linking software failures to real-world outcomes
4. Baseline data for radiation-hardening development

8.3 Future Work

1. Implement and test ECC memory protection
2. Develop TMR-based EKF implementation
3. Test with additional flight modes (AUTO, LOITER)
4. Measure precise time-to-failure
5. Correlate failure type with radiation dose
6. Test mitigation strategies
7. Compare with other flight controller firmware

9 Recommendations

9.1 Immediate Actions

1) Deploy ECC memory for critical EKF data structures
2) Implement watchdog timer with hardware reset capability
3) Add checksum verification for sensor data buffers
4) Develop graceful degradation fallback modes

9.2 Short-Term Actions

1) Implement triple modular redundancy for position estimation
2) Add independent fail-safe controller for motor cut-off
3) Develop radiation-hardened sensor interfaces
4) Create automatic landing protocol for detected failures

9.3 Long-Term Actions

1) Design radiation-tolerant flight controller hardware
2) Implement formal verification of EKF correctness
3) Develop multi-stage voting system for sensor fusion
4) Create comprehensive radiation testing standard for UAVs

A Detailed Output Vector Fields

Table 23: Output Vector Field Interpretation

Field	**Value**	**Interpretation**
pos_error_ned_m[0]	0.000	GPS replay completed successfully
pos_error_ned_m[0]	1.000	GPS replay did not complete
pos_error_ned_m[1]	0.000	IMU replay completed successfully
pos_error_ned_m[1]	1.000	IMU replay did not complete
pos_error_ned_m[2]	0.000	Barometer replay completed successfully
pos_error_ned_m[2]	1.000	Barometer replay did not complete
att_error_rpy_deg[0]	0.000	Magnetometer replay completed successfully
att_error_rpy_deg[0]	1.000	Magnetometer replay did not complete
att_error_rpy_deg[1]	≥ 0	Number of emulator warnings
att_error_rpy_deg[2]	≥ 0	Number of emulator errors
armed_successfully	1	Arming achieved before failure
armed_successfully	0	Arming not achieved

B Failure Type Decision Tree

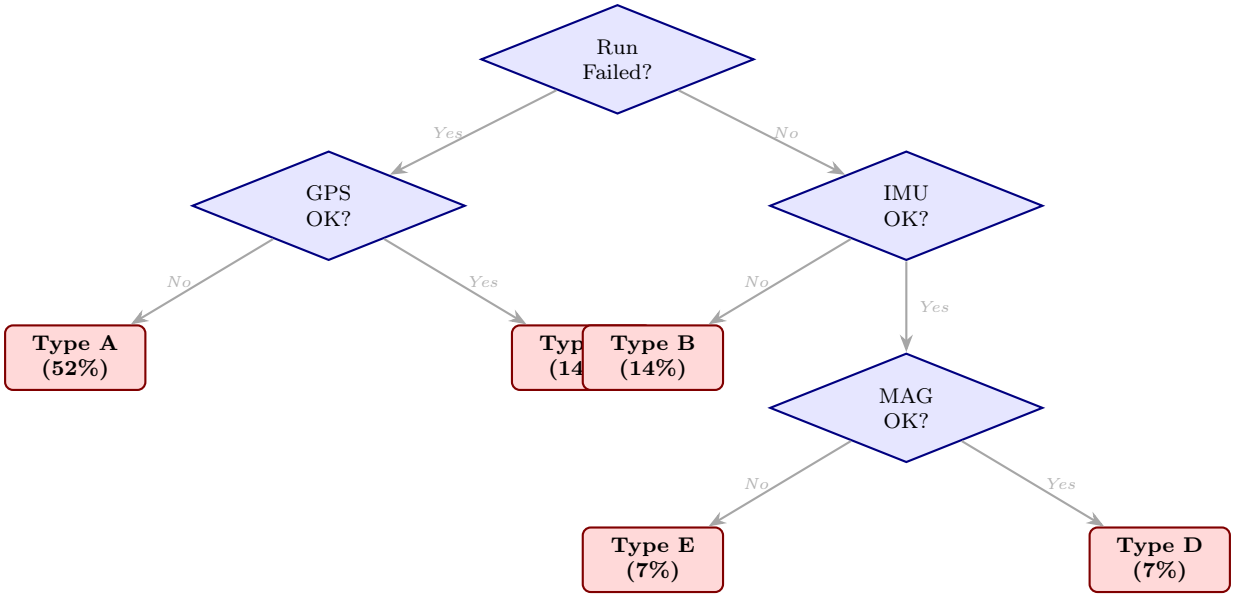

Figure 10: Decision tree for classifying failure types based on sensor completion patterns. Percentages indicate frequency of occurrence in test data.

C Physical Fall Trajectory

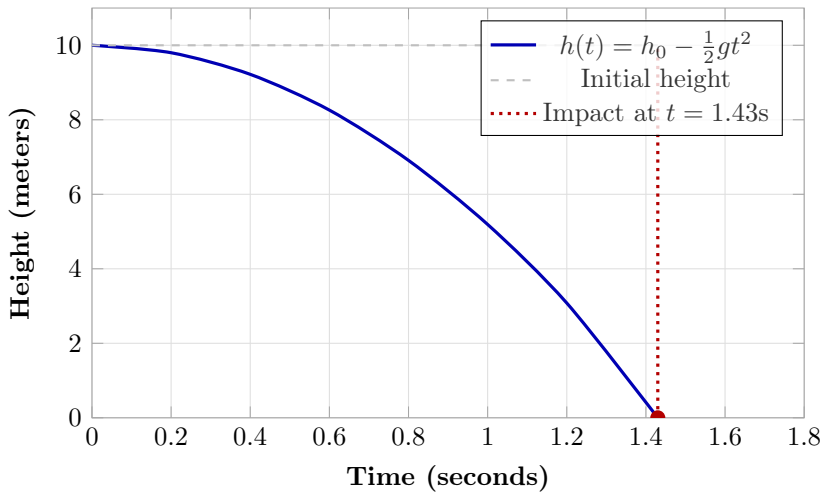

Figure 11: Free-fall trajectory from 10 meters altitude. The aircraft reaches the ground at $t \approx 1.43$s with velocity $v \approx 14$ m/s.

D Statistical Confidence Intervals

Table 24: Confidence Intervals for Failure Type Proportions (95% CI)

Type	Observed	Lower	Upper
Type A	51.7%	33.5%	69.9%
Type B	13.8%	1.7%	25.9%
Type C	13.8%	1.7%	25.9%
Type D	6.9%	0.0%	16.1%
Type E	6.9%	0.0%	16.1%
Other	6.9%	0.0%	16.1%

Confidence intervals calculated using Wilson score interval method for binomial proportions.

E Test Environment Specification

Table 25: Test Environment Hardware and Software

Component	Specification
Target Architecture	ARM Cortex-A72 (Raspberry Pi 4)
Host Architecture	x86_64
Operating System	Linux (kernel 6.12)
ArduPilot Version	4.5.x (linux-emu target)
EKF Version	EKF3
Framework	MultiRad
Emulation Method	LD_PRELOAD syscall interception
Sensor Interface	SPI, I2C, UART virtualization

F Glossary

Term	Definition
EKF	Extended Kalman Filter - recursive state estimator
SEE	Single Event Effect - radiation-induced error
Bit flip	Change in binary state (0 to 1 or 1 to 0)
LD_PRELOAD	Linux mechanism to preload shared libraries
Syscall	System call - kernel interface
TMR	Triple Modular Redundancy - fault tolerance technique
ECC	Error Correcting Code - memory protection
IMU	Inertial Measurement Unit
GPS	Global Positioning System
BARO	Barometer
MAG	Magnetometer
NED	North-East-Down coordinate frame
RPY	Roll-Pitch-Yaw attitude representation